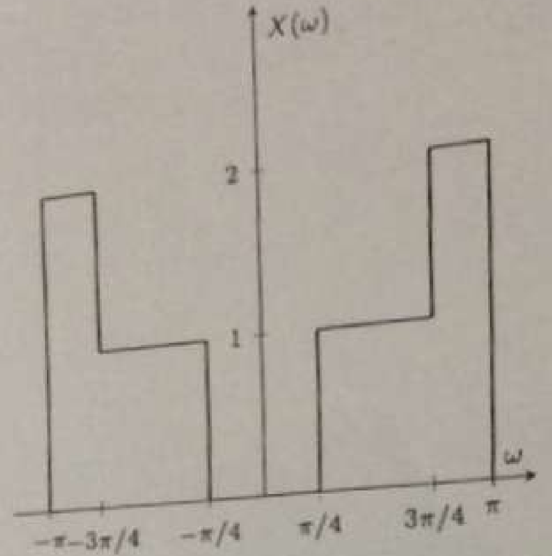

Figure 1: DTFT $X(\omega)$ for Problem 1 .

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Problems

1. (*Discrete-Time Fourier Transform*): Compute $x[n]$ for $X(\omega)$ as shown in Figure 1 . The final answer should be in terms of a real signal (without complex exponentials).

Hint: You can use the following Fourier Transform relationship:

$$\frac{\sin\left(\frac{W}{2}n\right)}{\pi n} \xleftrightarrow{\text{DTFT}} \operatorname{rect}\left(\frac{\omega}{W}\right)$$

2. (*Downsampling*): A discrete-time signal $x[n]$ has a Discrete-Time Fourier Transform (DTFT) $X(w)$ given by Figure 2.

$$2\pi \operatorname{rect}\left(\frac{n}{W}\right) \longleftrightarrow \frac{\sin\left(-\frac{W}{2}\omega\right)}{-\frac{W}{2}\omega}$$

$-\frac{W}{2}\omega$

$$2\pi \operatorname{rect}\left(\frac{n}{W}\right)^{2/3} \longleftrightarrow$$

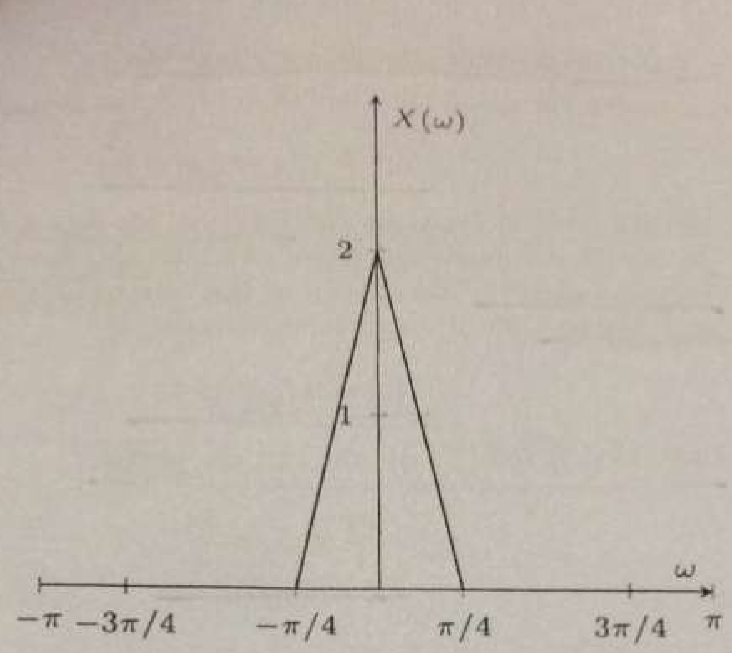

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Figure 2: DTFT of $x[n]$ for Problem 2

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Let

$$y[n] = x[2n]$$

be the downsampled signal (downsampling by a factor of 2). State the expression for $Y(\omega)$ in terms of $X(\omega)$, and plot it as a function of ω (do not derive the expression for $Y(\omega)$).

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3. (*Single Sideband Modulation*):

   (a) We know from the class lecture that the Complex Half-Band Filter has the impulse response given by

   $$h_{\mathrm{CHBF}}[n] = e^{j\frac{\pi}{2}n}\frac{\sin\left(\frac{\pi}{2}n\right)}{\pi n}.$$

   State $H_{\mathrm{CHBF}}(\omega)$, and draw its magnitude plot.

   (b) Again, from the class notes, we note that the Hilbert Transformer is given by

   $$h_{\mathrm{HT}}[n] = 2\frac{\sin\left(\frac{\pi}{2}n\right)}{\pi n}\sin\left(\frac{\pi}{2}n\right).$$

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State $H_{\mathrm{HT}}(\omega)$, and draw its magnitude plot.

(c) Consider the signal $x[n]$ from Problem 2. Let us define

$$\tilde{x}[n] \triangleq x[n] * h_{\mathrm{CHBF}}[n].$$

As discussed in class, we call $\tilde{x}[n]$ the complex analytic signal. In digital communications, we multiply $\tilde{x}[n]$ with the complex exponential $e^{j\omega_c n}$ (also known as the carrier) and then transmit its real part. Hence the transmitted signal is

$$y[n] = \mathrm{Re}\{e^{j\omega_c n}\tilde{x}[n]\}.$$

Draw $|\tilde{X}(\omega)|$ and $|Y(\omega)|$. You can assume that

$$\frac{2\pi}{4} < \omega_c < \frac{3\pi}{4}.$$